

ECO204: Project 1, Portfolios

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Introduction

This report provides a detailed analysis and construction of a one-month forward portfolio, incorporating U.S. 90-Day Treasury bills and a Synthetic Risky Asset (SRA) portfolio composed of Essex Property Trust Inc. (ESS) and Consolidated Edison Inc. (ED), targeted for October 2024. I utilized four methods to support portfolio analysis: Statistical Analysis, Shrinkage, Wiener Stochastic Process, and Ito Stochastic Process. Monthly returns data for the S&P 500, ESS, ED, and U.S. Treasury bills, covering the period from July 1994 to September 2024, were collected from various sources including the Centre for Research in Security Prices (CRSP), FRED, and Google Finance. The goal of this analysis is to evaluate the expected return and risk for each method and determine the best approach for constructing an optimal portfolio.

Executive Summary

The analysis reveals that the weights of ESS and ED in the portfolio vary depending on the method used, but ESS consistently receives a higher weight than ED. A higher weight means that a larger proportion of the total investment capital is allocated to ESS, indicating that it is viewed as a better asset for the portfolio across all methods. This is because ESS has a higher Sharpe Ratio, which means it offers better risk-adjusted returns.

The Sharpe Ratio measures how much additional return an investor earns for every unit of risk taken. A higher Sharpe Ratio indicates that an investment is more efficient, providing greater returns relative to its risk. ESS's higher Sharpe Ratio suggests that it is the most attractive asset in terms of balancing returns and risk, which is why it receives a larger share of the investment allocation.

In terms of risk and return of SRA, the Statistical Analysis method yielded the highest expected return, but it also carried the highest risk. Conversely, the Wiener Stochastic Process method struck a good balance between return and risk, offering the best risk-adjusted return. This makes the Wiener method the most effective for constructing the portfolio.

The report also introduces a value known as the threshold "c," which represents an individual's risk tolerance. The Wiener method had a threshold "c" value of 1.35, indicating that it offers a

balanced approach to risk and return compared to other methods, which showed either higher or lower risk levels.

Risk and Return Analysis

After generating the data, I conducted statistical calculation and summarized the results (shown in appendix A). Based on this calculation, I found the risk and return for both ESS and ED as well as the overall market indicated by S&P 500, and the risk free asset represented by the U.S. Treasury Bills.

	ESS	ED	S&P 500	US T-Bills
Risk	6.1%	5%	4.4%	0.2%
Return	1.3%	0.9%	0.8%	0.2%

Figure 1

As depicted in Figure 1 above, ESS is clearly riskier than ED and the S&P 500, with U.S. Treasury bills being the least risky. This risk pattern aligns with the returns, where the riskiest asset (ESS) yields the highest return (1.3%), while the least risky asset (U.S. Treasury bills) results the lowest return (0.2%). This demonstrates the general principle that higher risk is typically associated with higher returns, while lower risk correlates with lower returns.

Covariance Analysis

The covariance analysis of ESS, ED, S&P 500, and U.S. Treasury bills as shown in figure 2 provide insights into the relationship between these assets. The positive covariance among ESS, ED, and the S&P 500 suggests that these assets tend to move in the same direction, reflecting their sensitivity to broader market conditions. When the market performs well, these assets are likely to yield high returns, and conversely, when the market declines, they tend to decrease as well.

In contrast, the very low covariance values between U.S. Treasury bills and the other assets indicate that Treasury bills' returns are largely independent of market fluctuations. Regardless of

broader market conditions, Treasury bills tend to offer stable returns, making them a low-risk investment option.

Moreover, assets with high positive covariance increase portfolio risk when grouped together, while those with low or negative covariance, such as Treasury bills, can help reduce portfolio risk when paired with more volatile assets. This combination contributes to a more balanced portfolio with lower overall volatility, achieving a more stable expected return.

Var-Cov	ESS	ED	S&P 500	US T-Bills
ESS	0.0037652	0.0007614	0.0011900	0.0000048
ED	0.0007614	0.0025082	0.0003486	0.0000004
S&P 500	0.0011900	0.00034866	0.0019160	0.0000044
US T-Bills	0.0000048	0.0000004	0.0000044	0.0000033

Figure 2

Risk Premium and Sharpe Ratio

The risk premium and sharpe ratio are crucial metrics for assessing investment returns relative to risk. The risk premium measures the additional return an investor can expect to earn from holding a risky asset compared to a risk free asset, such as U.S Treasury bills in this case. Essentially, it reflects the additional return investors receive for taking on the extra risk associated with a particular investment. A higher risk premium indicates that the investment offers greater potential returns relative to the risk being taken.

The Sharpe Ratio is used to assess the risk-adjusted return of an investment. It is calculated by dividing the Risk Premium by the standard deviation (a measure of risk or volatility) of the asset's returns. The Sharpe Ratio provides insight into how much additional return an investor could earn for each unit of risk. A higher Sharpe Ratio indicates that the asset offers a better return for the level of risk involved.

	ESS	ED	S&P 500
Risk Premium	0.009	0.005	0.004
Sharpe Ratio	0.150	0.099	0.101

Figure 3

Based on my analysis shown in figure 3, I found that ESS has the highest Risk Premium compared to ED and the S&P 500, making ESS the more favorable asset in terms of return relative to risk. Furthermore, ESS also has the highest Sharpe Ratio, followed by the S&P 500, with ED showing the lowest Sharpe Ratio. This analysis suggests that ESS is the most attractive investment when considering risk-adjusted returns.

Optimal Synthetic Risky Asset (SRA)

To identify the optimal allocation for the Synthetic Risky Asset (SRA) portfolio, I calculated the risk and expected return for various weight combinations of Essex Property Trust Inc. (ESS) and Consolidated Edison Inc. (ED) under four different methods: Statistical Analysis, Shrinkage, Wiener Stochastic Process, and Ito Stochastic Process. Below is a summary of the key findings from these calculations.

	Weight ESS	Weight ED	Expected Return	SRA Risk
Stats	62%	38%	1.15%	0.14
Shrinkage	57%	43%	0.98%	0.05
Wiener	65%	35%	0.99%	0.05
Ito	69%	31%	0.84%	0.05

Figure 4

From the figure 4, we can observe the weight allocations for ESS and ED, along with their respective expected return and risk (SRA-Risk) under each method. The Stats method has the highest expected return, but it also comes with the highest risk . On the other hand, the

Shrinkage, Wiener, and Ito methods show significantly lower risks, but with varying expected returns.

Among these methods, the Wiener Stochastic Process achieves a balanced expected return while keeping the risk at a manageable level, making it the most attractive option in terms of risk-return trade-off. This allocation suggests a 65% investment in ESS and 35% in ED for the most optimal portfolio.

Next, I calculated the Sharpe Ratio and Risk Premium for the optimal SRA allocation under each method. These metrics help assess the risk-adjusted performance and the additional return expected from taking on risk, respectively.

	Optimal SRA Sharpe Ratio	Risk Premium
Stats	0.047	0.008
Shrinkage	0.116	0.006
Wiener	0.128	0.006
Ito	0.093	0.005

Figure 5

As seen in the table, the Wiener Stochastic Process has the highest Sharpe Ratio (0.1280), followed by Shrinkage (0.1161). This reinforces the previous conclusion that the Wiener method offers the best risk-adjusted return, making it the optimal choice for the SRA.

Threshold “c”

Next, I calculated the **threshold "c"** values for each method. The threshold "c" represents the level of market risk that an investor is willing to accept, and it helps determine the optimal portfolio choice based on an investor's risk tolerance.

Method	Stats	Shrinkage	Wiener	Ito
Threshold “c”	1.75	1.15	1.35	0.96

Figure 6

The figure above presents the threshold "c" values for each method. The Statistical method has the highest threshold, indicating a higher level of market risk. In comparison, the Wiener method has a threshold value, which represents a more balanced level of risk, suitable for typical market conditions. The Shrinkage method offers a threshold, which is lower than the Statistical method but still indicates a higher risk tolerance than the Wiener method. The Ito method presents the lowest threshold, suggesting a more conservative approach to risk, but less favorable in terms of return.

Recommendation

Based on the analysis, I recommend constructing the optimal portfolio by selecting the approach that best balances risk and return. The Wiener Stochastic Process method stands out as the most efficient option, offering a strong expected return while maintaining a manageable risk level. While methods like Statistical Analysis provide higher returns, they come with significantly greater risks, and the Shrinkage and Ito Stochastic Process methods reduce risk but also result in lower expected returns. Therefore, the Wiener method offers the best balance between risk and reward. I also suggest allocating a greater weight to Essex Property Trust (ESS) due to its superior risk-adjusted returns, as indicated by its higher Sharpe Ratio, and a smaller portion to Consolidated Edison (ED). This allocation provides a mix of strong growth potential and portfolio stability.

Conclusion

In conclusion, after evaluating the performance of each method for constructing the Synthetic Risky Asset (SRA) portfolio, the Wiener Stochastic Process method emerges as the most optimal choice. It offers a balanced portfolio with 65% allocated to ESS and 35% to ED, resulting in an expected return of 0.99% and an SRA risk of 0.05. With a Sharpe Ratio of 0.13, this method provides the best risk-adjusted return. The threshold "c" value of 1.35 further indicates that the portfolio's risk is within acceptable market levels. Therefore, the Wiener method is recommended for constructing an SRA portfolio that maximizes returns while effectively managing risk.

APPENDIX A

Introduction

This appendix provides supplementary tables mentioned in the main report. These visuals offer additional details and reinforce the data and analysis provided.

Statistics Summary

	DIS-RET	CLX-RET	SP500-RET	90-DAY US T-BILLS
Observations	363	363	363	363
Min	-0.215424	-0.153351	-0.169425	0.0000083
Arithmetic Mean	0.013091	0.008869	0.008307	0.0020034
Geometric Mean	0.011216185	0.007626673	0.007339495	0.0020018
Max	0.208134	0.208155	0.126844	0.0052311
AM - GM	0.001875	0.001242	0.000968	0.0000016
Sample Variance	0.003765161	0.00250815	0.001915997	0.0000033
Sample Std. Dev.	0.061360906	0.050081438	0.043772104	0.0018146
GM - Var/2	0.009333605	0.006372598	0.006381497	0.0020002

Variance - Covariance Matrix

Var-Cov	ESS	ED	S&P 500	US T-Bills
ESS	0.0037652	0.0007614	0.0011900	0.0000048
ED	0.0007614	0.0025082	0.0003486	0.0000004
S&P 500	0.0011900	0.00034866	0.0019160	0.0000044
US T-Bills	0.0000048	0.0000004	0.0000044	0.0000033

Returns vs Risk Chart



Portfolio Summary

	Stats	Shrinkage	Wiener	Ito
Weight ESS	62%	57%	65%	69%
Weight ED	38%	43%	35%	31%
Expected Return	1.15%	0.98%	0.99%	0.84%
Risk Premium	0.008	0.006	0.006	0.005
SRA Risk	0.14	0.05	0.05	0.05
SRA Sharpe Ratio	0.047	0.116	0.128	0.093
Threshold “c”	1.75	1.15	1.35	0.96

APPENDIX B

Detailed Explanation of Excel Workbook: Zahrani_Hanum_80340

Introduction

This appendix explains the steps I took to conduct my portfolio analysis in excel workbook Zahrani_Hanum_80340 as well as showing key formulas I used in each sheet.

Sheet 2: FRED 90-Day T-Bill Data

This sheet converts 90-Day T-Bill data from FRED into monthly yields spanning September 1981 to October 2024. Data was downloaded from FRED, converted from annualized yield to decimal format, and further broken down into semi-annual and monthly returns.

Key Formulas:

- Annual Return = Annualized Yield (%) / 100
- Semi-Annual Return = Annual Return / 2
- Monthly Return = $(1 + \text{Semi-Annual Return})^{(1/6)} - 1$

Sheet 3: CRSP-DATA

This sheet processes CRSP data for stocks ESS, ED, and the S&P 500 from July 1994 to December 2023. Data was imported, cleaned (invalid values removed), and common dates were aligned for comparative analysis.

Sheet 4: Google Finance-ESS

These sheets calculate monthly returns for ESS from January to October 2024 using daily data gathered through [google finance](#). Monthly returns were computed by dividing the closing price on the last day of the month by the first day's closing price, then subtracting it to 1.

Key Formulas:

- Google sheets → `GOOGLEFINANCE(ticker, [attribute], [start_date], [end_date|num_days], [interval])`

- $\text{Monthly Return} = (\text{Last Day Return} / \text{First Day Return}) - 1$

Sheet 5: Google Finance-ED

These sheets calculate monthly returns for ED from January to October 2024 using daily data gathered through [google finance](#). Monthly returns were computed by dividing the closing price on the last day of the month by the first day's closing price, then subtracting it to 1.

Key Formulas:

- Google sheets → `GOOGLEFINANCE(ticker, [attribute], [start_date], [end_date|num_days], [interval])`
- $\text{Monthly Return} = (\text{Last Day Return} / \text{First Day Return}) - 1$

Sheet 6: Raw Google

This sheet merges daily closing prices for ESS, ED, and the S&P 500 index from January to October 2024 gathered by [google finance](#). The data is organized for easier comparison of performance across different assets. Key dates, such as the first and last trading day of each month, are highlighted to aid in calculating monthly returns.

Key Formulas:

- Google sheets → `GOOGLEFINANCE(ticker, [attribute], [start_date], [end_date|num_days], [interval])`

Sheet 7: FILTERED-DATA

This sheet displays the monthly rates of return (RoR) for S&P 500, ESS, and ED. The data is filtered based on highlighted start and end prices for each month that I did in sheet 6. Monthly RoR values are then calculated and summarized for easy comparison.

Key Formulas:

- $\text{Rate of Return (RoR)} \rightarrow = (\text{End Price} / \text{Start Price}) - 1$

Sheet 8: MASTER-DATA

This sheet consolidates data for all assets, including ESS, ED, S&P 500, and U.S. Treasury Bills. The data is sourced from previous sheets, creating a comprehensive dataset for further analysis and modeling. U.S. Treasury Bills data is integrated using reference formulas to maintain consistency.

Key Formulas:

- Reference formula for U.S. Treasury Bills → ='Sheet Name'! Cell Number

Sheet 9: Summary-Stats

This sheet provides a statistical summary and charts of various assets' performance, including ESS, ED, S&P 500, and U.S. Treasury Bills. Key metrics like mean, variance, and Sharpe ratios are calculated, and visual representations are created to compare asset performance and risk.

To create the "Summary-Stats" sheet, I copied return data for ESS, ED, S&P 500, and U.S. Treasury Bills from the "MASTER-DATA" sheet and prepared a new table. I added 1 to each stock's return to facilitate the calculation of the geometric mean, which requires compounding returns. I then calculated basic statistics such as count, minimum, average, and maximum returns for each asset, as well as measures of risk like variance and standard deviation. The risk premium was determined as the difference between each stock's expected return and the risk-free rate, and the Sharpe ratio measured the risk-adjusted performance. To visualize these results, I created charts comparing risk and return metrics.

Key Formulas:

- Product of Values =PRODUCT(Number1:Number2)
- Count Values =COUNT(Value1:ValueX)
- Minimum Value =MIN(Number1:NumberX)
- Average (Mean) =AVERAGE(Number1:NumberX)
- Maximum Value =MAX(Number1:NumberX)
- Sample Variance =VAR.S(Number1:NumberX)
- Standard Deviation =SQRT(number)
- Covariance: =COVARIANCE.S(array1, array2)

- Geometric Mean = $(\text{Product of Stock x Returns})^{(1/\text{Total Observations})} - 1$
- Risk Premium = Expected Return of Risky Stock - Return of Risk-Free Stock
- Sharpe Ratio = Risk Premium / Risk

Sheet 10: Stats-Portfolio

The Stats-Portfolio approach uses a traditional optimization framework to find the best mix of two assets, ESS and ED, aiming to maximize the returns for a specified level of risk. This method relies on key statistics, including expected returns, variances (measuring each asset's risk), and covariances (indicating how the two assets move together). The primary goal is to create a portfolio that offers the highest possible return-to-risk ratio, quantified by the Sharpe ratio.

To perform the portfolio analysis using this method, I first gathered the necessary statistics: expected returns, variances, and covariances for ESS and ED. With these values, I calculated the optimal weights for each asset in the portfolio using formulas designed to achieve maximum returns for a given level of risk. After determining these weights, I computed the portfolio's expected return as a weighted average of the returns of ESS and ED, and I also calculated the portfolio's risk, taking into account both the individual risks of each asset and how they move together. Lastly, I evaluated the portfolio's Sharpe ratio by dividing the risk premium (the difference between the portfolio's expected return and the risk-free rate) by its risk, providing a measure of how well the portfolio compensates for the amount of risk taken.

Key Formulas:

- Portfolio Risk = $\text{SQRT}((\text{Weight1}^2 * \text{Var1}) + (\text{Weight2}^2 * \text{Var2}) + (2 * \text{Weight1} * \text{Weight2} * \text{Covariance}))$
- Optimal Weights Calculation:
 - $\text{Weight1} = (\text{RP1} * \text{Var2} - \text{RP2} * \text{Covariance}) / (\text{RP1} * (\text{Var2} - \text{Covariance}) + \text{RP2} * (\text{Var1} - \text{Covariance}))$
 - $\text{Weight2} = 1 - \text{Weight1}$
- Expected Return of SRA = $(\text{Weight1} * \text{Return1}) + (\text{Weight2} * \text{Return2})$
- Risk Premium = Expected Return of Risky Stock - Return of Risk-Free Stock

- Sharpe Ratio = Risk Premium / Risk
- $\text{Variance(SRA)} = (\text{Portfolio Risk})^2$
- $\text{Threshold C} = (\text{Return(SRA)} - \text{Risk-Free Return}) / (2 * \text{Variance(SRA)})$
- $\text{Optimal Beta} = (\text{Return(SRA)} - \text{Risk-Free Return}) / (2 * \text{C Value} * \text{Variance(SRA)})$

Sheet 11: Shrinkage-Portfolio

The Shrinkage-Portfolio method adjusts estimates for expected returns and risks by blending individual asset returns with broader market returns. This technique helps avoid overfitting and makes predictions more reliable, particularly when data is limited.

To implement the Shrinkage-Portfolio approach, I began by adjusting the expected returns for the assets ESS and ED. This was done by averaging their individual returns with the overall market return to reflect a more balanced estimate of performance. Next, I adjusted the risk of the assets using a shrinkage factor that blended the standard deviation of each asset with that of the broader market, providing a more robust measure of volatility. Using these adjusted values, I recalculated the portfolio's expected return and risk, determining the optimal weights for the assets based on maximizing returns relative to risk. Finally, I evaluated the portfolio's performance using the Sharpe ratio, which measures the return earned for each unit of risk taken.

Key Formulas:

- =SQRT(Value) — Used for calculating standard deviation from variance.
- $\text{Standard Deviation} = \text{SQRT(Variance)}$
- $\text{Adjusted Expected Return (Shrinkage)} = (\text{Stock Return} + \text{Market Return}) / 2$
- $\text{Adjusted Risk (Shrinkage)} = 1.3 * (\text{Std Dev of Stock} + \text{Std Dev of Market})$
- $\text{Portfolio Risk (SRA)} = (\text{Var_Stock1} * \text{Weight1}^2) + (\text{Var_Stock2} * \text{Weight2}^2) + (2 * \text{Weight1} * \text{Weight2} * \text{Covariance})$
- $\text{Expected Portfolio Return (SRA)} = (\text{Weight1} * \text{Average Return_Stock1}) + (\text{Weight2} * \text{Average Return_Stock2})$
- $\text{Risk Premium} = \text{Expected Return} - \text{Risk-Free Rate}$
- $\text{Weight1} = (\text{Risk Premium1} * \text{Var_Stock2} - \text{Risk Premium2} * \text{Covariance}) / (\text{Risk Premium1} * (\text{Var_Stock2} - \text{Covariance}) + \text{Risk Premium2} * (\text{Var_Stock1} - \text{Covariance}))$

- $\text{Weight2} = 1 - \text{Optimal Weight1}$
- $\text{Return(SRA)} = (\text{Optimal Weight1} * \text{Return1}) + (\text{Optimal Weight2} * \text{Return2})$
- $\text{Sharpe Ratio} = \text{Risk Premium} / \text{Risk}$
- $\text{Variance(SRA)} = (\text{Risk(SRA)})^2$
- $\text{Threshold C} = (\text{Return(SRA)} - \text{Risk-Free Return}) / (2 * \text{Variance(SRA)})$
- $\text{Optimal Beta} = (\text{Return(SRA)} - \text{Risk-Free Return}) / (2 * \text{C Value} * \text{Variance(SRA)})$

Sheet 12: Wiener Portfolio

The Wiener-Portfolio approach uses a stochastic process to model the random movement and volatility of asset returns over time. This method takes into consideration fluctuations and uncertainty inherent in financial markets, providing a dynamic and realistic view of how a portfolio might evolve in the future. The Wiener process is often used in finance to forecast and simulate paths of stock prices and other financial metrics, incorporating randomness to better capture market behavior.

To use the Wiener-Portfolio approach, I started by calculating the expected returns and risks for the assets ESS and ED while factoring in random market changes. This allowed me to estimate how prices and returns might move over time. Next, I determined the optimal mix of these assets in the portfolio to maximize return while managing risk. Using these adjusted values, I recalculated the expected return and overall risk of the portfolio. Finally, I evaluated how well the portfolio performs compared to the risk taken using the Sharpe ratio.

Key Formulas:

- $=\text{SQRT}(\text{number})$ — Calculating standard deviation from variance.
- $\text{Portfolio Risk (Wiener)} = (\text{Weight1}^2 * \text{Var1}) + (\text{Weight2}^2 * \text{Var2}) + (2 * \text{Weight1} * \text{Weight2} * \text{Covariance})$
- $\text{Expected Return (Wiener)} = (\text{Weight1} * \text{Return1}) + (\text{Weight2} * \text{Return2})$
- $\text{Risk Premium} = \text{Expected Return} - \text{Risk-Free Rate}$
- Optimal Weights Calculation (Weight1, Weight2) — Based on maximizing return and minimizing risk.
- $\text{Sharpe Ratio} = \text{Risk Premium} / \text{Portfolio Risk}$

- $\text{Variance(SRA)} = (\text{Risk(SRA)})^2$ — Calculation of portfolio variance.
- $\text{Threshold C for Beta} = (\text{Return(SRA)} - \text{Risk-Free Return}) / (2 * \text{Variance(SRA)})$
- $\text{Optimal Beta} = (\text{Return(SRA)} - \text{Risk-Free Return}) / (2 * \text{C Value} * \text{Variance(SRA)})$

Sheet 13: Ito-Portfolio

The Ito-Portfolio approach looks at how asset prices change continuously and how these changes impact the portfolio. By accounting for small, ongoing shifts in asset prices and their ups and downs, this method offers a detailed view of how a portfolio can perform over time. It considers both changes in volatility and how different assets move together.

To create and analyze the Ito-Portfolio, I began by modeling how the prices of the assets ESS and ED change continuously over time, focusing on capturing small and ongoing shifts in their values. This involved tracking both their expected returns and changes in risk, often referred to as volatility, to understand how much their prices could fluctuate. By accounting for how these assets might move together, I calculated the overall risk for the portfolio, considering both individual risks and their relationship with each other.

Next, I determined the best combination of ESS and ED to maximize the portfolio's returns while managing the level of risk. This involved finding the optimal weights for each asset that would yield the highest return relative to the risk taken. Once the expected returns and overall portfolio risk were calculated, I evaluated the portfolio's performance using the Sharpe ratio, which measures how much additional return is earned for each unit of risk taken. This comprehensive approach provided insight into how continuous price changes and varying risk levels affect the portfolio's behavior and potential returns.

Key Formulas:

- =SQRT(number) — Used to calculate the standard deviation, which represents risk.
- $\text{Portfolio Risk (Ito)} = (\text{Weight1}^2 * \text{Var1}) + (\text{Weight2}^2 * \text{Var2}) + (2 * \text{Weight1} * \text{Weight2} * \text{Covariance})$
- $\text{Expected Return (Ito)} = (\text{Weight1} * \text{Return1}) + (\text{Weight2} * \text{Return2})$
- $\text{Risk Premium} = \text{Expected Return} - \text{Risk-Free Rate}$

- Optimal Weights Calculation (Weight1, Weight2) — Adjusted to reflect continuous changes in prices and risks.
- Sharpe Ratio = Risk Premium / Portfolio Risk
- $\text{Variance(SRA)} = (\text{Risk(SRA)})^2$ — Used to calculate how much the portfolio's returns can vary.
- Threshold C for Beta = $(\text{Return(SRA)} - \text{Risk-Free Return}) / (2 * \text{Variance(SRA)})$
- Optimal Beta = $(\text{Return(SRA)} - \text{Risk-Free Return}) / (2 * \text{C Value} * \text{Variance(SRA)})$

Sheet 14: Portfolio Summary

The Portfolio Summary sheet brings together an overview of the portfolio's performance using various analysis methods, including Stats, Shrinkage, Wiener, and Ito. It highlights key metrics such as asset allocation, expected returns, risk levels, Sharpe ratios, and other indicators, offering a comprehensive view of how each method impacts the portfolio.

To create this summary, I consolidated key metrics from each of the different portfolio analysis methods using reference formulas like `=Sheet Name! Cell Number`. This allowed me to pull calculated values for asset weights, expected returns, risk premiums, overall portfolio risk, Sharpe ratios, and beta thresholds from their respective sheets. By organizing all the data in one structured table, it became easy to compare and analyze how each method affects portfolio performance.

Formulas and Functions Used:

- Reference formula → `=Sheet Name! Cell Number`

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